

NOTE 06 — MALWARE CASE STUDY

What is malware?

Malware means malicious software designed to perform unauthorized or harmful actions.

Examples include:

- Trojan
- Ransomware
- Spyware
- Worm
- Rootkit
- Keylogger
- Botnet malware
- Remote Access Trojan (RAT)

Fictional Malware Scenario

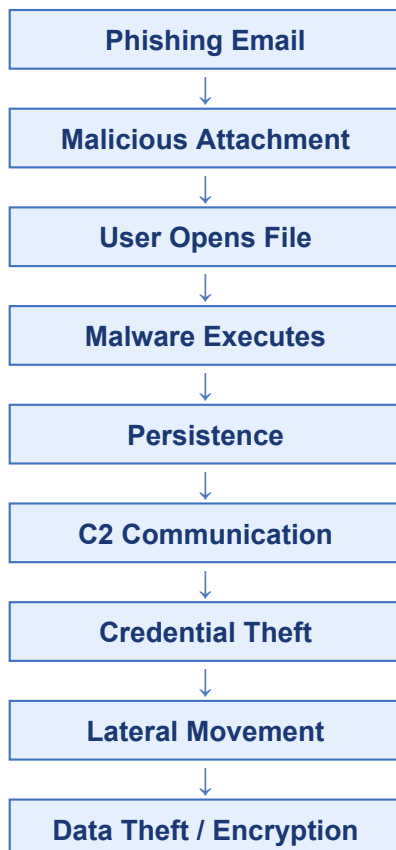
Scenario

An employee receives an email containing a malicious attachment.

The employee opens the attachment.

The malware executes.

A simplified attack chain could be:



Possible impact

1. Credential theft

The malware could attempt to steal credentials stored on or accessible from the endpoint.

2. Data theft

Sensitive documents could potentially be collected and exfiltrated.

3. Persistence

The attacker may attempt to maintain access after reboot.

4. Command and Control

The infected system may communicate with attacker-controlled infrastructure.

5. Lateral movement

The attacker may attempt to move from the infected endpoint to other systems.

6. Ransomware

Some malware can encrypt files and demand payment.

How can a SOC detect this?

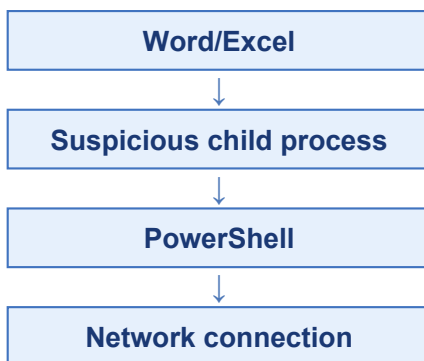
A SOC could investigate:

Email logs

Look for suspicious sender/domain/attachment.

EDR

Look for:



DNS logs

Look for suspicious domains or unusual DNS behavior.

Firewall/proxy logs

Identify outbound connections to suspicious destinations.

Authentication logs

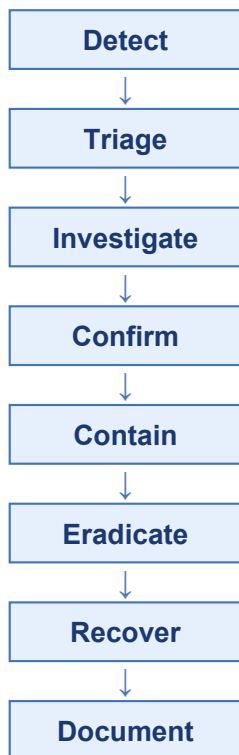
Look for unexpected logins.

SIEM

Correlate multiple events together.

SOC response

A possible response process:



Possible containment actions include:

- Isolate affected endpoint.
- Disable compromised account if appropriate.
- Block malicious indicators.
- Remove malicious software.
- Reset credentials.
- Search for additional affected systems.

Conclusion

Malware can affect confidentiality, integrity, and availability simultaneously.